

org-babel Maxima demo

Maxima Source Code Blocks in Org Mode

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Contents

NOTE

- to toggle image view: Ctrl-c Ctrl-x Ctrl-v
- to toggle L^AT_EX view: Ctrl-c Ctrl-x Ctrl-l
- to export as L^AT_EX file: Ctrl-c Ctrl-e l l
- to export as pdf file: Ctrl-c Ctrl-e l p
- to export as pdf file and view: Ctrl-c Ctrl-e l o

REFERENCES

Maxima References

- Maxima Manual
- Maxima by Example
- Symbolic Mathematics Using Maxima
- Graphics with MAXIMA

org-mode References

- Maxima Source Code Blocks in Org Mode

SOURCES

ob-maxima documentation site

- Maxima Source Code Blocks in Org Mode [main site]
- Maxima Source Code Blocks in Org Mode [mirror]
- Maxima Source Code Blocks in Org Mode [source]

Examples of Use

Calculator

The following source code block uses `maxima` as a calculator for powers of 12, where the powers are passed with a variable.

```
26.06280316745401
```

Solver

Of course, `maxima` is more than a calculator.

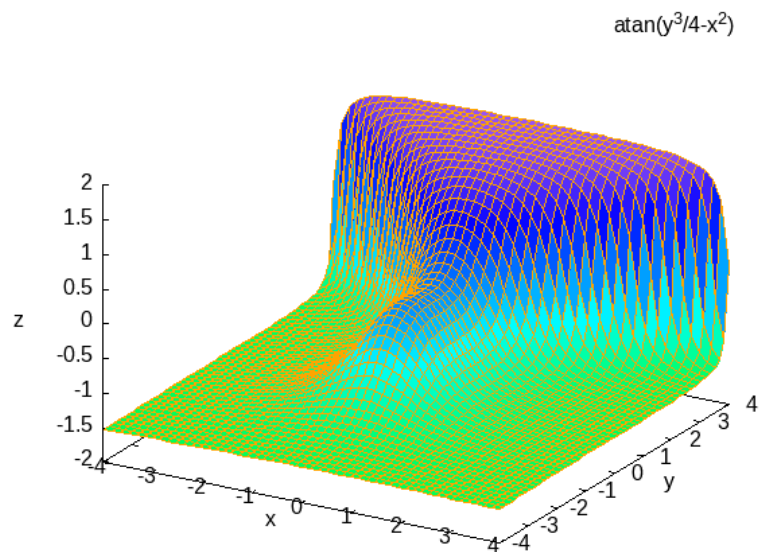
```
solve: solution:
```

```
x = - 4  
x = 4
```

```
[%t1, %t2]
```

3D plots

With `gnuplot` installed (4.0 or higher), 3D graphics are possible. This example is from a tutorial on the `maxima/gnuplot` interface.



Inline Display of Maxima $\text{\LaTeX}$ Output

Maxima code can be evaluated and displayed inline in Org mode through babel¹ as in the example below, based on RS initial example.

$$\frac{e^{-x}}{x}$$

$$\frac{e^{-x}}{x}$$

TODO Not Working Examples

TODO The :batch header argument

Setting the :batch header argument to batch:

- allows the use of the :lisp reader;
- provides a more verbose output;
- allows one to typeset calculations in $\text{\LaTeX}$.

¹(info “(org)Library of Babel”)

An example with the `:lisp` reader

Sample code block:

```
incorrect syntax: : is not a prefix operator
SpaceSpace:1
^
```

The first line of input computes the integral $\int_0^\infty t^z e^{-t} dt$, assuming that z is positive. The second input line uses the `:lisp` reader to print the internal representation of that result as a sexp.

An example with line numbering

By default, the command-line option `--very-quiet` is passed to the Maxima executable; this option suppresses the output of the start-up banner and input/output labels. That can be modified by setting `:cmdline --quiet`, which allows printing of input/output labels.

Sample code block:

Maxima's default is to print input in linear (or 1d) fashion, while output is printed in 2d.

LA_TE_X output

To produce **L**A_TE_X output for an extended computation, one needs to set-up a **L**A_TE_X printer. This example uses the `alt-display` package to do that. To print output as **L**A_TE_X, Maxima's 2d printer is replaced with `org_tex_display`; to ensure input lines are not echoed, its 1d printer is replaced with a sink, `org_no_display` (see also this Note).

Tangle this code block (`C-u C-c C-v t`):

Next, write a Maxima code block that sets the `:cmdline` header to read in the initialization code that was just tangled.

Maxima encountered a Lisp error: Condition in LOAD [or a callee]: INTERNAL-SIMPLE-FILE-ERROR: File error on #p"maxima-initialize.lisp": File does not exist. Automatically continuing. To enable the Lisp debugger set `debugger-hook` to nil. Condition in LOAD [or a callee]: INTERNAL-SIMPLE-ERROR: The tag RETURN-FROM-DEBUGGER is undefined. Broken at LOAD. Type :H for Help. 1 Maxima top-level 2 Maxima top-level 3 Return to top level. MAXIMA»

Note

Prior to version 5.47, Maxima could only pre-load a lisp file; to get around this constraint, the Maxima code is written into a lisp file, and the `#$` reader macro is used to read the Maxima code. In versions 5.47 and higher, the Maxima code can be put in a `.mac` file and pre-loaded without the need for such tricks.

The `:graphics-pkg` header argument

The `:graphics-pkg` header argument can be set to use either Maxima's built-in `plot` package (the default), or the `draw` package.

TODO The `plot` package

The `plot` package is the default package that provides a simplified interface to `gnuplot`. Here is an example that creates a `gif` file.

```
plot2d( sin(x), [x,0,2*%pi]);

images/ob-maxima-plot.gif
```

The `draw` package

The `draw` package has more features than `plot`, including an object-oriented interface to several graphics engines, including `gnuplot`. Here is an example that creates an `svg` file containing the graph of a discontinuous function.

```
f(x) := if x>0 then cos(x) else if x<0 then 0;
draw2d(
  line_width=2, grid=true, yrange=[-1.2,1.2],
  explicit(f,x,0,%pi), explicit(f,x,-1,0),
  fill_color=white, ellipse(0,0,0.05,0.05,0,360),
  fill_color=blue , ellipse(0,1,0.05,0.05,0,360));

[width=.9]images/ob-maxima-draw
```

- Note Internally, `ob-maxima` uses the function `set_draw_defaults`. Because this function overwrites the existing defaults, using it in code blocks with the `:graphics-pkg draw` header argument will cause the source block evaluation to fail silently.

Additional Notes

Toggle inline display of latex code

Latex code in org mode can be displayed inline by 'C-c C-x C-l'. To remove the inline display 'C-c C-c' is used. This is described further in the manual².

Set scale of output

If the inline display of the equations are illegible, the scale can be set by customising the variable 'org-format-latex-options', by setting the :scale variable to a value >1.

Export

This exports nicely to both html (C-c C-e h h) and pdf (C-c C-e l p). See³ and⁴ in the manual.

Noweb expansion

NOTE: I have not tested this yet, but as Eric Schulte noted on the mailing list: “Alternately, if you really want to get fancy you could use noweb expansion⁵ to insert the results of the imaxima code block into a latex code block, and then use the existing latex code block functionality to convert the imaxima output to images of different types depending on the export target.”⁶

²(info “(org)Previewing L^AT_EX fragments”)

³(info “(org)Exporting code blocks”)

⁴(info “(org)The export dispatcher”)

⁵(info “(org)noweb”)

⁶Re: imaxima babel